Any communication system must have three basic units: transmitter, receiver and transmission line. In natural voice communication, when we speak, our mouth acts as the transmitter and when we hear, our ear acts as the receiver. In between these two, air acts as the transmission line.

Natural voice communication has a few unique characteristics, as given below.

- It is instantaneous, with virtually zero delay between the transmitter and receiver. Thus, it is also real-time communication.
- This type of communication can be best described as “When I hear you, I feel you and I see you,” and so “When I speak with you, you feel me and you see me.” Thus, natural voice communication is more than just hearing (voice). It is seeing (video) and feeling (perception), too.
- It supports man-mobility. As we move, we can speak, see and feel, too.
- It is interactive.
- It is one where identity of the transmitter and receiver never differs from place to place. For example, your name does not change as you move from one place to another.
- It is fully wireless.

Natural voice communication is not without its limitations. It is essentially a short-haul form of communication. Its coverage area is small. Using this mode, communication beyond a few metres is not possible.

Many scientific and technological developments are aimed to overcome such challenges.

The telegraph: early long-distance text communication

Ancient civilisations of China, Greece and Egypt, for the purpose of long-distance communication, used drumbeats or smoke signals for exchange of information. As an improvement to such primitive systems, the semaphore was developed in early 1700s. In the semaphore system, a series of hilltops was used as movable arms to signal letters and numbers. Two telescopes were used to see the other stations. Later, improvements were made only in terms of increasing the coverage area of communication.

A Morse key, circa 1900 (Credit: upload.wikimedia.org)

These ancient systems were susceptible to failures of line-of-sight between communicating stations, low visibility and bad weather, among others.

Subsequently, various other forms of communication were developed for long-distance communication to exchange information, ideas, thoughts, etc. A few examples are letters, books and newspapers.

Over the years, many systems of exchanges have been developed—including postal systems and newspapers/books delivery systems—basically as man-managed systems. All these developments attempted to achieve long-distance communication but could not attain any unique characteristic of natural voice communication. However, in 1790s came the opportunity to attain the desired goal with the invention of the first electrical communication, called the telegraph. On March 2, 1791, the first telegraphic message was sent.

Telegraphy is made of two words: ‘tele’ meaning at a distance and ‘graphy’ meaning to write. With this, long-distance communication of text messages was conceptualised. In fact, this was an improved version of a semaphore.

Discovery of radio communication paved the way for developing electrical telegraphy. Using semaphore networks, early telegraphy was carried out by French scientist Claude Chappe.

The credit for commercial telegraphy goes to Samuel Morse. A code known as Morse code was developed around 1837 for signalling of alphabets. Accordingly, text messages, as a series of codes, were transmitted between the transmitter and receiver using electrical signals.

Telegraph as the first long-haul communication system got systemised in postal communication. Post offices at a distance served as transmitters and receivers. But telegraphy, unlike natural voice communication, did not have features such as interactive, instantaneous, fully wireless, man-mobile, unique addressed and multi-mode communication. It connected only two geographical points at a time for service.

Electrical communication that started its journey with the telegraph is still marching ahead in multiple advanced forms.

Long-distance voice communication developed over the years

The next long-distance electrical communication is known as the telephone. The word telephone is made up of two words: ‘tele’ meaning at a distance and ‘phone’ implying sound. In a telephone, interactive voice communication is made over a long distance in real time (zero latency). In this system, telephone handsets act as transmitters/receivers (transceivers). These connect geographical points for the purpose of communication without any support for mobility.

All handsets have a microphone and an earphone. The microphone converts voice signal into electrical signal, and acts as the transmitter. Whereas, the earphone acts as the receiver. Electrical signals get transmitted over either wired or wireless mode, between two phones connected at a distance.

Alexander Graham Bell invented the telephone in 1876. The first call he made was to his assistant on March 10, 1876. His first words over his invention were, “Mr Watson, come here. I want to see you.”